AutoRoadSAM: Multimodal Remote Sensing Road Extraction With Structure-Semantic Awareness via Auto-Prompting Vision Foundation Models

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***Abstract*—The integration of multimodal data holds great promise for advancing road extraction in remote sensing. However, existing approaches are limited by the lack of unified end-to-end frameworks for diverse modality combinations, suboptimal multimodal feature fusion, and challenges in capturing the slender, winding, and complex topological structures of roads. In this article, we propose AutoRoadSAM, a novel end-to-end framework for multimodal road extraction that fully exploits the powerful visual representation capabilities of the segment anything model (SAM) and, for the first time, introduces an auto-prompting mechanism via a dynamic snake convolution-based decoder. This decoder adaptively generates task-specific prompts by capturing fine-grained local geometric features from auxiliary modality branches, enabling precise alignment with complex road structures. To further enhance multimodal feature fusion and topological perception, we design the cross-modal information interaction (CMII) module, which facilitates global context modeling and cross-modal interaction, while strengthening the representation of intricate road topology through multidirectional snake scanning. Moreover, we incorporate a mask decoder with cross-polarity-aware linear attention (CPLAM) to boost decoding efficiency and effectively address pixel imbalance. Together, these innovations enable AutoRoadSAM to achieve superior structure- and semantic-aware road extraction across diverse modality combinations. Extensive experiments on six public datasets and four modality combinations demonstrate that AutoRoadSAM consistently outperforms state-of-the-art methods, validating the effectiveness and generalization capability of each proposed component. The code is available at https://github.com/NWPUFranklee/AutoRoadSAM.git**

***Index Terms*—Auto-prompting, feature fusion, multimodal remote sensing, road extraction, vision foundation models.**

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I. Introduction

THE spatial distribution of roads is a key indicator of the structural layout and locational attributes of a region, playing a crucial role in urban planning [1], [2], transportation management [3], [4], and national defense infrastructure. Traditional road extraction methods [5], [6], [7] predominantly rely on identifying road features from optical remote sensing imagery. However, with the maturation of deep learning techniques, the accuracy of these approaches has reached a performance plateau. The rapid development of sensor technology has enabled the acquisition of a wide variety of multimodal remote sensing data. Consequently, researchers [8], [9], [10] have increasingly incorporated features from multiple modalities into road extraction tasks, achieving notable improvements. For instance, digital surface model (DSM) [11] data provide essential 3-D spatial cues; thermal infrared (TIR) [12] features help suppress interference from heat-emitting objects; trajectory data [13], such as GPS-derived vehicle traces, capture the actual spatial distribution of roads; and synthetic aperture radar (SAR) [14] imagery offers robust visual information under adverse weather conditions.

In recent years, numerous deep learning-based methods [15], [16], [17] have been developed for road extraction from remote sensing imagery. Compared to traditional approaches based on handcrafted features [18], [19], [20], these methods have substantially improved extraction accuracy, largely owing to the widespread adoption of convolutional neural networks (CNNs) [21], transformer [22], and Mamba framework [23]. While CNNs excel at capturing fine-grained road features, transformers and Mamba architectures are effective at modeling long-range contextual dependencies in single-modality scenarios [24], [25], [26]. However, these approaches are generally restricted to single-modality input, limiting their

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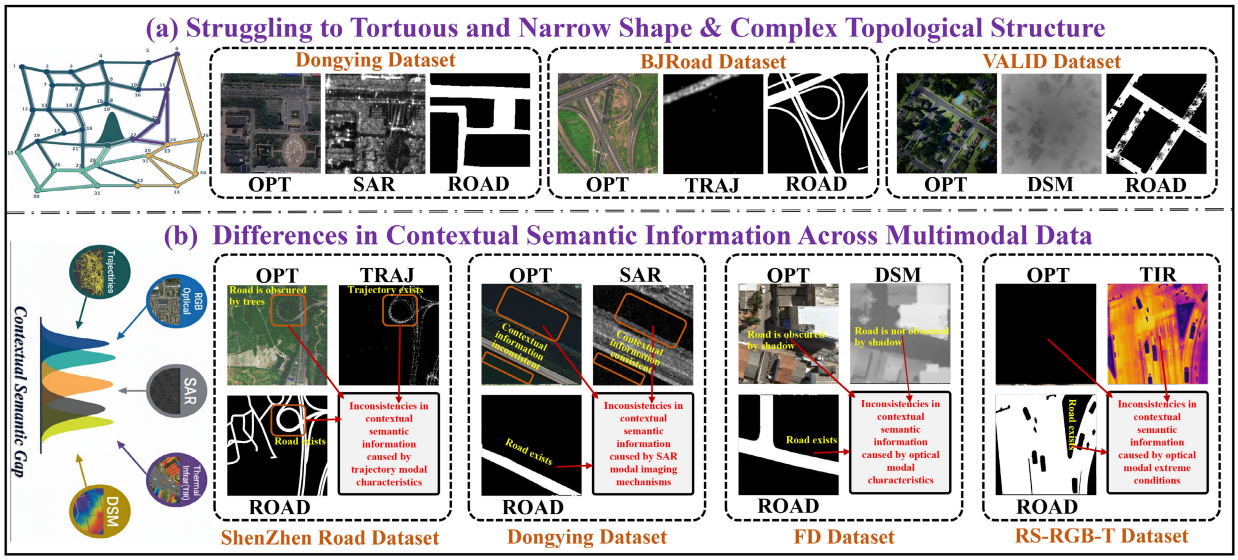


Fig. 1. Representative challenges in multimodal road extraction. (a) Intricate and highly connected road topologies with slender and tortuous structures hinder robust segmentation. (b) Substantial semantic inconsistencies and ambiguities across modalities, caused by occlusions, spectral similarities, and incomplete information, pose significant obstacles to unified feature learning and generalizable extraction.

applicability in multimodal contexts. To address this limitation, researchers have proposed a range of multimodal road extraction methods. For example, Li et al. [27] introduced DF-DRUNet to improve the fusion of optical images and GPS trajectories via a gated fusion module, while Liu et al. [28] developed CMMPNet, which leverages dual autoencoders and enhancement modules to dynamically propagate complementary information among modalities. Nevertheless, these methods typically support only two or three modalities and exhibit limited generalization across diverse modalities, posing significant challenges in large-scale remote sensing interpretation applications.

Given these challenges, we turned to the segment anything model (SAM) [29], a vision foundation model capable of zero-shot segmentation through user-provided prompts and trained on large-scale datasets. Although SAM has been applied to multimodal remote sensing semantic segmentation tasks [30], [31], [32], its application to multimodal road extraction remains largely unexplored. This is primarily due to the unique and complex challenges faced in this domain, as shown in Fig. 1. First, severe pixel imbalance presents a fundamental challenge [see Fig. 1(a)]. Different from natural scene segmentation, where object pixels are often more balanced, in multimodal road extraction, roads typically occupy only a small fraction of the image, resulting in a significant foreground-background imbalance. Second, inconsistencies in contextual semantic information across modalities further complicate the task [see Fig. 1(b)]. For instance, in optical images, roads may be occluded by trees or shadows; in TIR images, roads and backgrounds may exhibit similar thermal signatures; DSM data can suffer from inconsistent elevation cues due to vegetation cover; trajectory data may contain missing or abnormal paths; and SAR images often present high visual similarity between roads and other linear features like rivers. These discrepancies create a substantial semantic gap across modalities, making unified segmentation especially challenging when multiple modalities are involved. Finally, the unique geometric properties of roads, their slender, winding, and networked topological structures, pose additional difficulties. Current transformer-based architectures, including SAM, are not specifically designed to capture such elongated and complex structures, resulting in suboptimal performance in multimodal road extraction tasks.

To address these challenges, we propose AutoRoadSAM, a novel end-to-end auto-prompting framework for multimodal road extraction. At its core, AutoRoadSAM incorporates the cross-modal information interaction (CMII) module within the feature encoder, built upon the VMamba architecture [33]. By introducing local constraints and facilitating information interaction, CMII enables comprehensive global contextual understanding across modalities and effectively learns the topological structure of road networks. Moreover, it promotes the modeling of interobject relationships within each modality. To further enhance the feature representation capability, we design the snake auto-prompt decoder (SAPD), which utilizes dynamic snake-shaped convolutions [34] and leverages auxiliary modalities during pixel-level decoding. This not only realizes self-prompting functionality, eliminating the need for manual input, but also allows the network to adaptively capture the tortuous and elongated geometric features typical of roads. In addition, we introduce the mask decoder with cross-polarity-aware linear attention (CPLAM), integrating a linear attention mechanism [35] into the decoding stage. This mechanism allows AutoRoadSAM to focus on both the interactions among road and background classes, as well as the influence of pixel polarity, thereby alleviating the negative impact of severe class imbalance on segmentation accuracy. The main contributions of this work are summarized as follows.

1) We propose AutoRoadSAM, an end-to-end auto-prompt framework for multimodal remote sensing road extraction, eliminating the need for manual prompts and demonstrating strong robustness and generalization across diverse modality combinations.

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2) We design the CMII based on VMamba architecture, which enables both local alignment and global modeling of multimodal features, facilitating accurate topological structure learning and effective context interaction.
3) We develop the SAPD, which integrates dynamic snake-shaped convolutions and auxiliary modality guidance for adaptive pixel-level prompt generation and robust learning of tortuous, elongated road geometries.
4) We introduce the CPLAM, which incorporates CPLAM into the mask decoder, effectively alleviating severe pixel imbalance and improving accuracy in complex multimodal scenarios.

The article is structured as follows. Section II provides an overview of the related works. Section III introduces the AutoRoadSAM framework and its associated components. The experimental results and discussion analysis are presented in Sections IV. Finally, the conclusion is drawn in Section V.

II. Related Works

A. Unimodal Road Extraction

Unimodal road extraction methods, which focus on single-source remote sensing imagery, have witnessed substantial progress in recent years through the development of novel network architectures and advanced context modeling techniques. GCB-Net [36] addresses the challenge of modeling global spatial relationships by incorporating a global context awareness module, thereby effectively integrating global features for improved road segmentation. To mitigate background interference from objects such as railways that exhibit similar linear structures to roads, the C^2Net framework [37] introduces a global–local context perception module to capture comprehensive topological features. CoANet [6] is designed to enhance road topological connectivity in satellite imagery challenged by occlusions and complex backgrounds. This approach leverages a strip convolution module and a connectivity attention module to jointly learn segmentation and pairwise pixel dependencies, effectively improving the integrity of extracted road networks. DDU-Net [5] tackles the difficulty of extracting narrow and elongated road structures, especially under vegetation shadow occlusion, by incorporating a dilated convolution attention module to expand the receptive field and enhance multiscale feature representation. To address the issue of dense occlusions caused by buildings and trees, OARENet [38] introduces a road occlusion-aware module that explicitly models road texture features and strengthens road connectivity. RADANet [7] employs a deformable attention module to better capture long-range dependencies and contextual spatial correlations, thereby enhancing the extraction of complex road topologies. However, unimodal methods cannot fully exploit the complementary information provided by heterogeneous data sources, resulting in limited robustness and generalization in challenging real-world environments.

B. Multimodal Road Extraction

Multimodal road extraction methods leverage complementary information from heterogeneous data sources, such as optical, LiDAR, SAR, and trajectory data, to address the limitations inherent in single-modality approaches. DeepDualMapper framework [39] utilizes a gated fusion module to effectively combine information from multiple modalities and employs a densely supervised refinement decoder to generate high-precision road maps in a coarse-to-fine manner. To mitigate the discontinuities caused by occlusions from buildings and vegetation in high-resolution images, DECCFNet [40] adopts a dual-encoder architecture, fusing HRI and LiDAR data at both channel and spatial dimensions via a cross-modal feature fusion module, and incorporates a multidirectional strip convolution to better capture road morphology. UFNet [41] addresses the challenge of road discontinuities and omissions by integrating RGB imagery and LiDAR-derived surface models (DSM, DTM, and nDSM), thus providing complementary geometric cues and achieving strong performance on suburban datasets. LCIRE-Net [42] learns modality-specific features through dual encoders, dynamically interacts information via the attention mechanism, fuses cross-modal features with multiscale learning, and employs feature fusion for precise edge refinement, thereby enabling continuous and accurate road extraction even under severe occlusions. To overcome the complexity and redundancy of traditional multimodal fusion strategies, CMIPNet [43] streamlines the integration of GPS trajectories or LiDAR data with optical imagery, enhancing fusion efficiency while reducing unnecessary information. DEFNet [44] introduces a dual-layer evidence fusion network that adaptively fuses pixel-level and superpixel-level cues through a cross-attention feature interaction module, addressing the inflexibility of conventional fusion mechanisms. Despite these advances, current multimodal methods often face challenges such as modality misalignment, semantic inconsistency, and cross-modal complementarities, indicating the need for more robust and adaptive strategies.

C. SAM-Based Road Extraction

Recent advances in foundation models, particularly the SAM [29], have inspired a series of studies exploring their potential for remote sensing road extraction through task adaptation and prompting strategies. SAM-Road framework [45] addresses the challenge of large-scale, vectorized road network extraction by reframing the task as dense semantic segmentation and fine-tuning the SAM image encoder to better fit satellite imagery. Road-SAM [46] enhances the efficiency of road extraction from large, high-resolution remote sensing images by introducing adapters and explicit visual prompts, thereby customizing the general SAM model for road segmentation tasks. The SAM-embedding-based method [47] focuses on pixel-level road damage extraction, overcoming challenges such as blurry boundaries, varying damage sizes, and uneven spatial distributions through the integration of SAM embeddings. To address the limitations of the general SAM model in handling complex backgrounds and the reliance on manual prompts, TPP-SAM [13] leverages trajectory points as prompts to enable zero-shot road extraction from high-resolution remote sensing images. Despite their promising performance, SAM-based approaches are still constrained by the pretraining domain gap, the need for effective adaptation to

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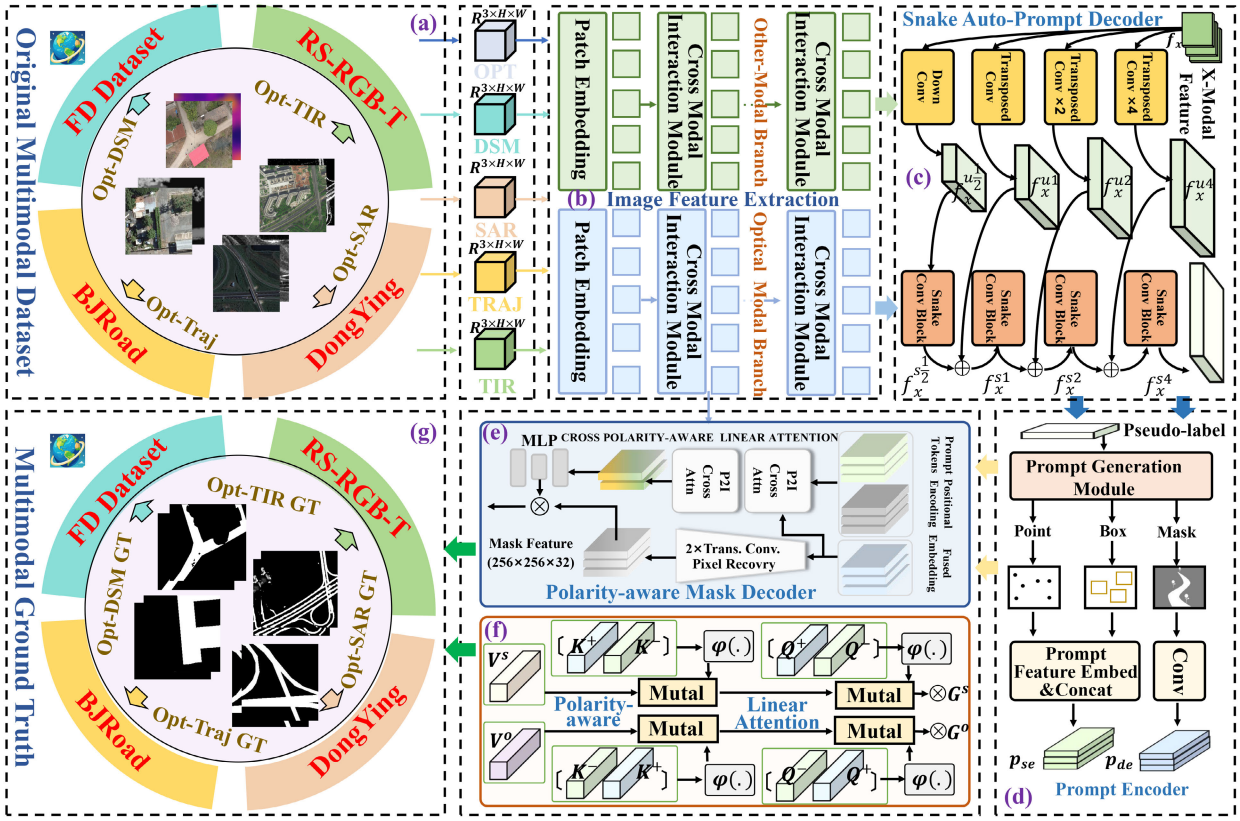


Fig. 2. Overall architecture of the proposed AutoRoadSAM framework. (a) Multimodal remote sensing datasets and diverse ground truths. (b) Feature extraction backbone with CMII modules for heterogeneous multimodal fusion. (c) SAPD for hierarchical prompt generation and decoding guided by auxiliary modalities. (d) Prompt encoder, consistent with the original SAM design, for encoding point, box, and mask prompts. (e) Mask decoder with CPLAM for polarity-aware mask prediction. (f) Details of the CPLAM mechanism. (g) Visualization of multimodal ground truth maps.

remote sensing characteristics, and the challenge of leveraging cross-modal cues for further improvement.

III. Methodology

In this section, as illustrated in Fig. 2, we present a comprehensive overview of the proposed AutoRoadSAM framework. The architecture is composed of four major components: 1) feature extraction backbone centered on the CMII module, which enables effective fusion of heterogeneous multimodal features; 2) auto-prompting architecture, termed the SAPD, which leverages auxiliary modality features for hierarchical prompt generation and decoding; 3) prompt encoder, consistent with the original SAM design, responsible for encoding primitive point, box, and mask prompts; and 4) mask decoder with CPLAM, which performs polarity decoding based on primary modality features to generate high-fidelity road masks.

A. AutoRoadSAM Framework Structure

The overall architecture of the proposed AutoRoadSAM framework is designed to enable efficient and robust multimodal road extraction by explicitly modeling cross-modal interactions, auto-prompting, and polarity-aware decoding, as illustrated in Fig. 2. The core of feature encoder is the CMII module, denoted as $\varphi_{\text{CMII}}(\cdot,\cdot)$, which receives paired inputs from four possible modality combinations, where each pair consists of $\boldsymbol{I}_x \in \mathbb{R}^{3\times H\times W}$ and $\boldsymbol{I}_y \in \mathbb{R}^{3\times H\times W}$, where I_x is designated as the primary modality and I_y as the auxiliary modality, with H and W representing the image height and width, respectively. The CMII is constructed based on the Mamba architecture and aims to extract global interaction features $\boldsymbol{f}_k \in \mathbb{R}^{256\times 64\times 64}$ for $k \in \{x, y\}$. These features effectively capture cross-modal dependencies and reduce semantic discrepancies, thereby enabling robust multimodal fusion and laying the foundation for subsequent auto-prompting and decoding. The process is formulated as

$$\boldsymbol{f}_x, \boldsymbol{f}_y = \varphi_{\text{CMII}}\left(\boldsymbol{I}_x, \boldsymbol{I}_y\right). \tag{1}$$

As shown in Fig. 2(c), to enable AutoRoadSAM to operate independently of external user prompts while retaining the strong visual representation capabilities imparted by the prompt-based strategy in SAM, we introduce a SAPD, denoted as $\varphi_{\text{SAPD}}(\cdot)$. The auxiliary branch feature $\boldsymbol{f}_y$, which contains both intramodal and fused cross-modal information, is fed into the SAPD to generate a pseudo-label $\boldsymbol{p}_{\text{gt}} \in \mathbb{R}^{\text{cls}\times H\times W}$, where cls denotes the number of semantic classes. This process is given by

$$\boldsymbol{p}_{\text{gt}} = \varphi_{\text{SAPD}}\left(f_y\right). \tag{2}$$

Subsequently, as illustrated in Fig. 2(d), the pseudo-label $\boldsymbol{p}_{\text{gt}}$ is passed to the prompt encoder $\varphi_{\text{PE}}(\cdot)$, which outputs sparse prompt embeddings $\boldsymbol{p}_{\text{se}}$ (representing point and box prompts) and dense prompt embeddings $\boldsymbol{p}_{\text{de}}$ (representing

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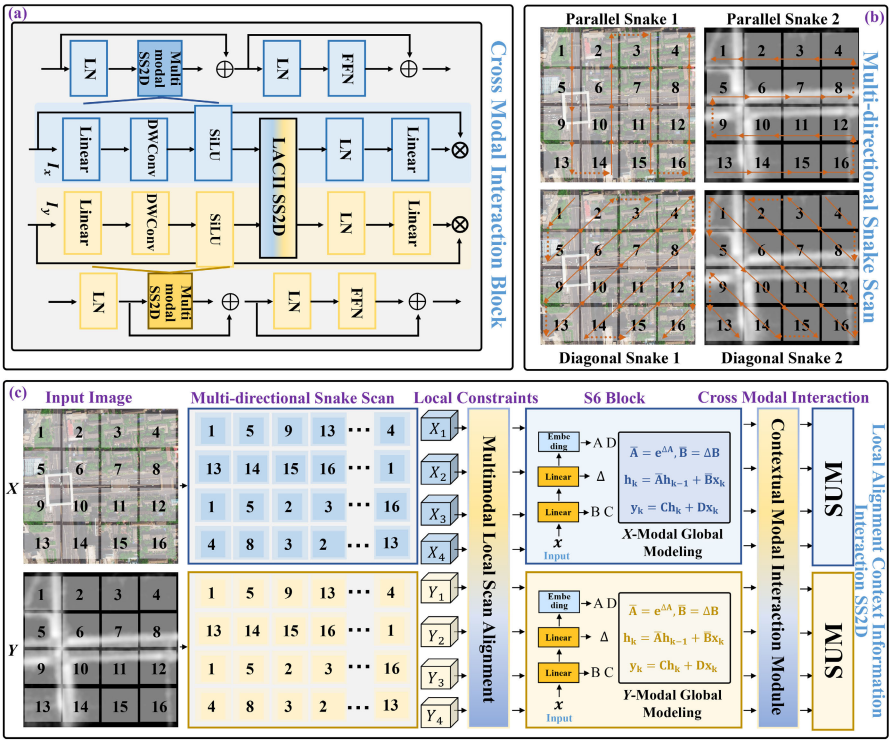


Fig. 3. Detailed illustration of the proposed CMII module and its core components. (a) Overall structure of the CMII module, which incorporates the multimodal SS2D and LACII modules for cross-modal feature fusion. (b) Four-directional snake scan strategy in the LACII module, including two parallel and two diagonal paths, designed to capture both orientation continuity and fine-grained topological features of roads. (c) End-to-end workflow of the LACII module, showing multidirectional snake scan, MLSA, global modeling via the S6 block, and contextual modal interaction for adaptive multimodal feature integration.

mask prompts). These embeddings effectively encode both the spatial location and category information of the object region

$$\boldsymbol{p}_{\text{se}}, \boldsymbol{p}_{\text{de}} = \boldsymbol{\varphi}_{\text{PE}}\left(\boldsymbol{p}_{\text{gt}}\right). \tag{3}$$

Finally, as shown in Fig. 2(e), to address pixel imbalance and enhance discriminability between road and nonroad regions, the polarity-aware mask decoder, denoted as $\boldsymbol{\varphi}_{\text{CPLAM}}(\cdot)$, takes as input the primary modal global feature $\boldsymbol{f}_x$ along with the prompt embeddings $\boldsymbol{p}_{\text{se}}$ and $\boldsymbol{p}_{\text{de}}$, and outputs the final segmentation mask $\boldsymbol{P}_{\mathcal{GT}}$ as follows:

$$\boldsymbol{P}_{\mathcal{GT}} = \boldsymbol{\varphi}_{\text{CPLAM}}\left(\boldsymbol{f}_x, \boldsymbol{p}_{\text{se}}, \boldsymbol{p}_{\text{de}}\right). \tag{4}$$

The above pipeline ensures that AutoRoadSAM can automatically and effectively leverage multimodal cues and self-generated prompts for accurate and robust road extraction from complex remote sensing data.

B. CMII Module

The CMII module is designed as the core component of the feature extraction stage, aiming to achieve both effective contextual modeling within each modality and robust integration of contextual information across modalities. The specific structure is shown in Fig. 3(a). This dual capability enables the learning of global interactive features that are mutually aware and fully exploit complementary information from heterogeneous sources. In conventional multimodal feature extraction, self-attention mechanisms are widely adopted but often encounter limitations in capturing the unique topological properties of road networks and efficiently modeling cross-modal interactions [48]. To address these challenges, the proposed CMII replaces the self-attention block with a multimodal 2-D selective scan (SS2D) block, as $\boldsymbol{\varphi}_{\text{MSS2DB}}(\cdot)$, while maintaining the overall architecture consistent with the original transformer block. Specifically, the CMII leverages SS2D-based contextual modeling, a feedforward neural network, and residual connections. The feature transformation process for modality k is formulated as follows:

$$\mathbf{Z}_k = \boldsymbol{\varphi}_{\text{MSS2DB}}\left(\boldsymbol{\varphi}_{\text{LN}}\left(\mathbf{X}_k\right)\right) + \mathbf{X}_k \tag{5}$$

$$\mathbf{X}_k^{\text{out}} = \boldsymbol{\varphi}_{\text{FFN}}\left(\boldsymbol{\varphi}_{\text{LN}}\left(\mathbf{Z}_k\right)\right) + \mathbf{Z}_k \tag{6}$$

where $\mathbf{X}_k$ is the input feature of modality k, $\boldsymbol{\varphi}_{\text{LN}}(\cdot)$ denotes the LayerNorm operation, $\boldsymbol{\varphi}_{\text{FFN}}(\cdot)$ is the feedforward neural network, $\mathbf{Z}_k$ is the intermediate feature, and $\mathbf{X}_k^{\text{out}}$ represents the output feature. The detailed operation within the multimodal SS2D block is further calculated as

$$\mathbf{X}_k^{\text{out}} = \boldsymbol{\varphi}_{\text{LL}}\left(\boldsymbol{\varphi}_{\text{LACII}}\left(\boldsymbol{\varphi}_{\text{LDS}}\left(\mathbf{X}_{k_1}, \mathbf{X}_{k_2}\right)\right)\right) \otimes \boldsymbol{\varphi}_S\left(\mathbf{X}_{\setminus k}\right) \tag{7}$$

where $\boldsymbol{\varphi}_{\text{LDS}}(\cdot)$ consists of a linear layer, depth-wise convolution, and SiLU activation; $\boldsymbol{\varphi}_{\text{LACII}}(\cdot,\cdot)$ denotes the parameter-shared local alignment context information interaction (LACII) module for multimodal fusion; $\boldsymbol{\varphi}_{\text{LL}}(\cdot)$ is a linear layer followed by LayerNorm; $\boldsymbol{\varphi}_S(\cdot)$ denotes the SiLU activation function; and $\mathbf{X}_{\setminus k}$ represents the auxiliary modality feature. Different from generic state-space or sparse attention mechanisms, the proposed CMII module is specifically designed to capture

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cross-modal and multiscale contextual dependencies tailored for the structural characteristics of road networks.

C. Local Alignment Context Information Interaction

The LACII module with SS2D constitutes the core of CMII, as shown in Fig. 3(b), designed with three major capabilities: local feature alignment, global context modeling, and dynamic multimodal feature interaction. To effectively address the complex topology and fine-grained texture variations of multimodal road features, we propose an enhanced snake scan strategy, as shown in Fig. 3(c), denoted as $\varphi_{\mathrm{SS}}(\cdot)$, which extends the original SS2D scanning strategy [49]. For each input modality feature $\mathbf{X}_k$, two parallel and two diagonal snake scan paths are employed, ensuring both the continuity of semantic information along roads and the capture of orientation-related texture features. This multidirectional scanning further supports comprehensive modeling of the diverse topological structures in subsequent processing. The scanning process can be calculated as follows:

$$\boldsymbol{S}_k^c = \boldsymbol{\varphi}_{\mathrm{SS}}\left(\mathbf{X}_k\right), \quad c \in \{1,2,3,4\} \tag{8}$$

where c indexes the four scan directions. After obtaining the scanning features $\boldsymbol{S}_k^c$, the original VMamba framework directly feeds them into the S6 block $\boldsymbol{\varphi}_{\boldsymbol{S6}}(\cdot)$ for global modeling. However, as road extraction is challenged by local discrepancies, both between different road segments and between roads and background, direct fusion may be suboptimal. To address this, we introduce a local multimodal scan alignment module $\boldsymbol{\varphi}_{\mathrm{LMSA}}(\cdot)$, which performs local alignment on the multimodal scan features from each direction, as features $\bar{\mathbf{S}}_k^c$

$$\bar{\mathbf{S}}_k^c, \bar{\mathbf{S}}_{\backslash k}^c = \varphi_{\mathrm{LMSA}}\left(\boldsymbol{S}_k^c, \boldsymbol{S}_{\backslash k}^c\right), \quad c \in \{1,2,3,4\}. \tag{9}$$

These aligned features are then processed through the S6 block for contextual global modeling

$$\mathbf{Z}_k^c = \boldsymbol{\varphi}_{S6}\left(\bar{\mathbf{S}}_k^c\right). \tag{10}$$

Since the globally aligned features at this stage do not yet encode cross-modal semantic consistency, we further apply a contextual modal interaction module (CMIM) $\varphi_{\mathrm{CMI}}(\cdot)$ to the aligned global features $\mathbf{Z}_k^c$ from each direction before feature merging. This operation dynamically reduces modality discrepancies and fuses multimodal information, resulting in directionally fused features $\bar{\mathbf{Z}}_k^c$ as follows:

$$\bar{\mathbf{Z}}_k^c, \bar{\mathbf{Z}}_{\backslash k}^c = \boldsymbol{\varphi}_{\mathrm{CMI}}\left(\mathbf{Z}_k^c, \mathbf{Z}_{\backslash k}^c\right), \quad c \in \{1,2,3,4\}. \tag{11}$$

Finally, the directionally fused features across all scan directions are aggregated to produce the output feature

$$\mathbf{X}_k^{\mathrm{out}} = \sum_{c=1}^{4} \bar{\mathbf{Z}}_k^c. \tag{12}$$

The LACII module leverages multidirectional snake scan, local multimodal alignment, and dynamic context interaction to deliver robust and topology-aware feature representation for complex road extraction scenarios. Distinct from linear state-space and axial/strip attention, LACII introduces adaptive local connectivity modeling to explicitly enhance the continuity and geometric alignment of fine-grained road structures.

D. MLSA Module

The multimodal local scan alignment (MLSA) module is designed to address the significant local variations in feature distributions among different modalities, which pose a major challenge for subsequent multimodal feature integration. As shown in Fig. 3(b), to achieve effective local alignment, MLSA seeks a minimum-cost transport plan between multimodal feature distributions within the LACII framework. Given local feature matrices $\mathbf{S}_k^c = \left[\mathbf{s}_{k,1}^c; \ldots; \mathbf{s}_{k,N}^c\right] \in \mathbb{R}^{N\times V}$ and $\mathbf{S}_{\backslash k}^c = \left[\mathbf{s}_{\backslash k,1}^c; \ldots; \mathbf{s}_{\backslash k,N}^c\right] \in \mathbb{R}^{N\times V}$ from the primary modality k and auxiliary modality $\backslash k$ along scan direction c, we first construct the pairwise cost matrix $\mathbf{C} \in \mathbb{R}^{N\times N}$ using cosine distance

$$\mathbf{C}(i,j) = 1 - \frac{\mathbf{s}_{k,i}^c \cdot \mathbf{s}_{\backslash k,j}^c}{\left\|\mathbf{s}_{k,i}^c\right\|_2 \left\|\mathbf{s}_{k,j}^c\right\|_2}. \tag{13}$$

Let $a \in \mathbb{R}^N$ and $b \in \mathbb{R}^N$ denote the marginal distributions of features, typically set as uniform, i.e., $a_i = b_j = 1/N$ for all i, j. The optimal transport plan $\mathbf{P}^* \in \mathbb{R}^{N\times N}$ is obtained by minimizing the transport cost with entropic regularization

$$\min_{\mathbf{P}\geq 0} \ \mathcal{L}_{\mathrm{OT}}\left(\mathcal{C}_{\mathrm{total}}(\mathbf{P}) - \epsilon H(\mathbf{P})\right).$$

The total transport cost is defined as

$$\mathcal{C}_{\mathrm{total}}(\mathbf{P}) = \sum_{i=1}^{N}\sum_{j=1}^{N} \mathbf{C}(i,j)\,\mathbf{P}(i,j) \tag{14}$$

and the entropic regularization term is

$$H(\mathbf{P}) = -\sum_{i=1}^{N}\sum_{j=1}^{N} \mathbf{P}(i,j)\log \mathbf{P}(i,j) \tag{15}$$

subject to the marginal constraints

$$\sum_{j=1}^{N} \mathbf{P}(i,j) = a_i, \quad \sum_{i=1}^{N} \mathbf{P}(i,j) = b_j \ \forall i,j \tag{16}$$

where $\epsilon > 0$ is the regularization parameter, and a, b are typically set as uniform distributions, i.e., $a_i = b_j = 1/N$. This problem can be efficiently solved by the Sinkhorn–Knopp algorithm. Specifically, let $K = \exp(-\mathbf{C}/\epsilon)$ denote the Gibbs kernel. The optimal $\mathbf{P}^*$ is computed via iterative updates

$$u^{(t+1)} = \frac{a}{Kv^{(t)}}, \quad v^{(t+1)} = \frac{b}{K^\top u^{(t+1)}} \tag{17}$$

where $u, v \in \mathbb{R}^N$, and / denotes element-wise division. After convergence, the transport plan is given by

$$\mathbf{P}^* = \mathrm{diag}(u)\ K\,\mathrm{diag}(v). \tag{18}$$

The aligned feature representations are then calculated as

$$\bar{\mathbf{S}}_k^c = \mathbf{P}^*\mathbf{S}_k^c, \quad \bar{\mathbf{S}}_{\backslash k}^c = (\mathbf{P}^*)^\top \mathbf{S}_{\backslash k}^c. \tag{19}$$

Through this theoretically grounded and differentiable alignment based on optimal transport [50], MLSA effectively reduces local semantic discrepancies between modalities, facilitating robust and discriminative multimodal feature fusion.

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E. Contextual Modal Interaction Module

The CMIM serves as the key component for multimodal feature interaction and fusion, as shown in Fig. 3(b), enabling the effective integration of complementary information from multiple modalities. Following global feature modeling, CMIM dynamically allocates weights to each modality, allowing semantic information from different sources to be adaptively fused based on their contextual relevance. Specifically, for each direction c, the aligned global features $\mathbf{Z}_k^c$ and $\mathbf{Z}_{\setminus k}^c$ are first concatenated and passed through a multilayer perceptron (MLP) to learn the joint multimodal characteristics. The resulting feature is then activated by a ReLU function $\Phi(\cdot)$ followed by a Sigmoid function $\sigma(\cdot)$, which suppresses irrelevant responses and assigns a normalized weight coefficient in the range $[0, 1]$ to each modality as follows:

$$\mathbf{w}_k = \sigma\left(\mathrm{MLP}\left(\mathrm{Cat}\left[\mathbf{Z}_k^c, \mathbf{Z}_{\setminus k}^c\right]\right)\right). \tag{20}$$

The learned weights are subsequently used to linearly combine the features of the two modalities, yielding the directionally fused representation $\bar{\mathbf{Z}}_k^c$ as follows:

$$\bar{\mathbf{Z}}_k^c = \mathbf{w}_k\, \mathbf{Z}_k^c + (1 - \mathbf{w}_k)\, \mathbf{Z}_{\setminus k}^c. \tag{21}$$

Through this adaptive weighting and fusion scheme, the CMIM effectively suppresses interference from irrelevant or noisy modalities while enhancing the contribution of discriminative multimodal features, thereby providing robust support for subsequent road extraction tasks.

F. Snake Auto-Prompt Decoder

The SAPD is designed to effectively reconstruct road structures by capturing their elongated, tubular morphology and complex topological relationships, which are often inadequately represented by conventional convolutional operations. Drawing inspiration from the encoder–decoder architecture [51] and MFNet [52], this decoder leverages the deformable and direction-aware properties of snake convolutions for enhanced morphological feature extraction in multimodal scenarios. As shown in Fig. 2(c), initially, a series of transposed convolution layers $\varphi_{\mathrm{T.conv}}^i(\cdot)$ are applied to the input auxiliary features $\mathbf{X}_y$, progressively restoring spatial resolution at multiple scales (e.g., $\times 2$, $\times 4$), and generating hierarchical multiscale feature maps $\mathbf{T}_i$. To further enrich the scale representation, a downsampling convolution $\varphi_{\mathrm{D.conv}}^i(\cdot)$ is employed to obtain lower resolution features. The overall process is calculated as

$$\mathbf{T}_i = \begin{cases} \varphi_{\mathrm{T.conv}}^i\left(\mathbf{X}_y\right), & i \in \{2, 3, 4\} \\ \varphi_{\mathrm{D.conv}}^i\left(\mathbf{X}_y\right), & i = 1. \end{cases} \tag{22}$$

Subsequently, inspired by the iterative decoding mechanism of UNet, each scale-specific feature $\mathbf{T}_i$ is processed through a serpentine convolution module $\varphi_{\mathrm{S.conv}}^i(\cdot)$ to better capture the fine-grained contours and orientation of road targets. The resulting features are then upsampled via $\varphi_{\mathrm{Up}}^2(\cdot)$ by a factor of two, and integrated with the features from the adjacent higher resolution level through a residual connection. This iterative refinement is described by

$$\mathbf{R}_{i+1} = \varphi_{\mathrm{Up}}^2\left(\varphi_{\mathrm{S.conv}}^i\left(\mathbf{T}_i\right)\right) + \mathbf{R}_i. \tag{23}$$

After obtaining the pseudo-label mask from SAPD, sparse prompts are generated by extracting keypoints and bounding boxes along the predicted road structures, while dense prompts are directly derived from the mask's connected regions.

G. Polarity-Aware Mask Decoder

The mask decoder with CPLAM serves as a core solution for addressing pixel-level class imbalance in road extraction tasks. As shown in Fig. 2(e) and (f), the proposed CPLAM is specifically designed to overcome the inherent limitation of standard linear attention, which tends to ignore negative feature values. This enhancement enables the model to learn distinctive background representations, which are crucial for accurately distinguishing road regions from complex backgrounds. We first decompose the input features via a learnable linear projection $\mathbf{W}_L$ and a ReLU activation, separating the positive and negative components for both queries and keys. The process is calculated as follows:

$$\mathbf{L}^+ = \Phi\left(\mathbf{W}_L \mathbf{X}_k\right), \quad L \in \{Q, K\} \tag{24}$$

$$\mathbf{L}^- = \Phi\left(-\mathbf{W}_L \mathbf{X}_k\right), \quad L \in \{Q, K\} \tag{25}$$

where $\Phi(\cdot)$ denotes the ReLU activation function, and $\mathbf{L}^+$ and $\mathbf{L}^-$ represent the positive and negative components, respectively. Furthermore, we reconstruct the semantic roles of queries, keys, and values by concatenation and splitting operations, yielding symbol correlation and dissimilarity features

$$\mathbf{Q}_{\mathrm{sim}} = \mathrm{Cat}\left[\mathbf{Q}^+, \mathbf{Q}^-\right] \tag{26}$$

$$\mathbf{Q}_{\mathrm{opp}} = \mathrm{Cat}\left[\mathbf{Q}^-, \mathbf{Q}^+\right] \tag{27}$$

$$\mathbf{K} = \mathrm{Cat}\left[\mathbf{K}^+, \mathbf{K}^-\right]. \tag{28}$$

This enables the attention mechanism to explicitly model interactions among positive-positive, negative-negative, positive-negative, and negative-positive feature pairs. The symbol-aware attention $\mathbf{Att}_{\mathrm{sim}}$, symbol-agnostic attention $\mathbf{Att}_{\mathrm{opp}}$, and the final global polarity-aware attention $\mathbf{Att}$ are obtained by the linear attention and concatenation, as follows:

$$\mathbf{Att}_L = \frac{\mathbf{Q}_L\left(\mathbf{K}^{\mathrm{T}}\mathbf{V}\right)\mathbf{G}_L}{\mathbf{Q}_L\left(\frac{1}{V}\sum_j \mathbf{K}_j\right)^{\mathrm{T}}}, \quad L \in \{\mathrm{sim}, \mathrm{opp}\} \tag{29}$$

$$\mathbf{Att} = \mathrm{Cat}\left[\mathbf{Att}_{\mathrm{sim}}, \mathbf{Att}_{\mathrm{opp}}\right] \tag{30}$$

where $\mathbf{G}_L$ denotes the polarity-aware coefficient matrices, which are optimized to capture the complementary relationships between same-sign and opposite-sign feature interactions. By introducing polarity-awareness into the linear attention mechanism, the CPLAM enables the decoder to simultaneously leverage both foreground and background cues, thereby enhancing the ability to address pixel imbalance and improving the precision of road extraction.

H. Loss Function

To effectively address the foreground-background class imbalance inherent in road extraction tasks, we employ the Dice loss as the sole objective for end-to-end optimization. Dice loss is particularly suitable for segmentation scenarios where the foreground (road) pixels are significantly sparser

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TABLE I

QUANTITATIVE COMPARISON OF AUTOROADSAM AND 16 STATE-OF-THE-ART METHODS ON THE BJROAD, SHENZHEN ROAD, AND FD DATASETS IN TERMS OF OA, MF1, MIOU, AND KAPPA. RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modality	Methods	BJRoad				Shenzhen Road				FD Dataset			
		OA	mF1	mIoU	Kappa	OA	mF1	mIoU	Kappa	OA	mF1	mIoU	Kappa
UniModal	UNet	71.38	64.18	59.42	70.05	64.40	53.39	50.59	56.99	83.40	80.00	73.72	80.66
	OARENet	74.87	74.06	62.62	74.54	72.09	65.66	52.27	68.47	91.68	81.52	74.98	82.33
	C^2Net	76.53	67.62	60.73	76.38	67.77	57.91	51.27	61.79	94.04	80.37	79.00	80.48
	RoadCT	79.68	70.42	60.91	75.36	66.89	59.43	53.76	63.47	88.92	82.74	79.23	84.96
	Road-SAM	82.55	70.33	62.80	74.50	71.74	60.90	55.01	66.35	93.47	85.35	81.62	86.77
Multimodal	DF-BSFNet	81.94	71.86	61.64	81.57	72.70	61.61	56.20	65.71	86.63	83.30	82.30	84.93
	DECCFNet	80.72	71.46	63.89	77.35	68.97	62.15	54.63	64.18	91.38	85.92	81.47	89.26
	CMIPNet	80.47	70.35	64.18	76.89	69.84	63.27	55.91	65.73	92.15	86.47	80.32	88.94
	LCIRE-Net	78.94	69.64	62.57	77.46	69.38	61.38	53.78	62.83	96.62	83.36	79.41	89.81
	MFNet	80.15	72.98	65.19	77.14	71.41	65.34	56.00	68.24	90.41	87.14	79.65	87.67
	FlexiSAM	83.12	74.65	64.37	78.94	70.85	64.12	57.46	66.83	92.74	88.03	85.15	90.86
	FTransUNet	83.87	76.54	65.16	79.80	70.21	64.38	57.03	65.95	95.34	90.00	83.52	90.57
	MultiSenseSeg	83.35	77.07	66.17	77.59	69.68	62.96	56.43	66.38	89.45	86.90	85.49	88.92
	SACFNet	84.36	74.11	63.44	79.10	70.65	67.73	55.81	65.77	92.63	89.83	86.01	90.78
	DEFNet	83.41	75.50	64.18	77.94	71.10	65.47	57.51	66.92	93.49	85.55	84.45	89.68
	RingMo-SAM	86.13	75.32	65.07	80.37	71.32	64.21	58.05	70.12	93.89	86.77	85.19	91.09
	AutoRoadSAM	85.64	77.29	67.83	81.23	73.10	66.33	59.72	69.10	95.03	91.29	87.81	93.12

than the background. Given the predicted probability map $\mathbf{P}$ and the ground-truth mask $\mathbf{Y}$, the Dice loss is formulated as

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i p_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} p_i + \epsilon} \tag{31}$$

where p_i and y_i denote the predicted probability and ground-truth label for pixel i, N is the total number of pixels, and ϵ is a small constant for numerical stability. This loss is applied to the output of the mask decoder, ensuring that the network is explicitly optimized for overlap between the predicted and reference road masks.

IV. EXPERIMENTS AND ANALYSIS

A. Dataset Description

1) Shenzhen Road Dataset [53]: The Shenzhen Road dataset comprises high-resolution remote sensing imagery collected from Shenzhen, Guangdong, China. It contains 614 training pairs and 152 test pairs, each consisting of optical and trajectory modality with a resolution of 1024 × 1024 pixels. The label space is binary, distinguishing only between "road" and "background." The ground sampling distance is 0.5 m, with an overall coverage of approximately 2000 km^2. The trajectory modality is constructed from extensive GPS data sampled from taxis and buses, including timestamp, longitude, latitude, speed, heading, and vehicle state flags.

2) BJRoad Dataset [54]: The BJRoad dataset features both optical and a rare trajectory modality, the latter aggregated from over 50 million trajectories recorded by 28 000 vehicles (predominantly taxis) in Beijing, China. The dataset includes 1350 registered optical-trajectory image pairs (1024 × 1024 pixels), covering roughly 100 km^2. All pairs are annotated as "road" and "other." During training, images are cropped into 256 × 256 patches, resulting in 21 600 pairs, split as 70% for training, 10% for validation, and 20% for testing.

3) FD Dataset [41]: The FD Dataset consists of high-resolution (20 cm) imagery from Area 01 of the Federal District, Brazil, characterized by dense road networks with significant occlusion from vegetation. Two modalities are provided: optical and DSM, though only DSM is used in our experiments. Each GeoTIFF is 3034 × 3122 pixels. The dataset is partitioned into 5675, 1008, and 1044 tiles for training, validation, and testing, respectively.

4) VALID Dataset [55]: The VALID dataset is synthetically generated using Unreal Engine 4, encompassing six distinct urban and rural scenarios (e.g., airport, skyscraper district, villa, European street, seaside town, and snowy mountain). Each scene is rendered under diverse weather and lighting conditions and captured at altitudes of 20, 50, and 100 m. All images and corresponding DSMs are 1024 × 1024 pixels with pixel-level annotations. While the original annotation covers 31 categories, only the "road" label is retained for evaluation, with all other classes set to background. The dataset is split as 50% for training, 16.7% for validation, and 33.3% for testing.

5) Dongying Dataset [56]: The Dongying dataset comprises 7831 pairs of optical and SAR images (256 × 256 pixels, 1-m GSD) acquired over Dongying, Shandong, China, using Gaofen-2 (optical) and Gaofen-3 (SAR) satellites. The original dataset contains five categories (road, building, water, farmland, and vegetation), but only the "road" class is retained (others set to background). The dataset is divided equally into training and testing sets, as in the original work. Unlabeled pixels, originally set to 255, are mapped to 5 for training but are excluded from evaluation metrics.

6) RS-RGB-T Dataset [57]: The RS-RGB-T dataset contains 4024 registered RGB and thermal image pairs (640 × 512 pixels) acquired by uncrewed aerial vehicles (UAVs) over urban road scenes, with 2514 images captured during daytime and 1510 at night. Pixel-level annotations are provided for nine categories, but only the "road" class is preserved in our experiments. The data is split into training, validation, and test sets in a 7:1:2 ratio, maintaining consistent proportions of daytime and nighttime scenes across subsets.

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TABLE II

COMPARISON OF APLS AND TOPO SCORES FOR ALL COMPETING METHODS ACROSS SIX BENCHMARK DATASETS. RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modality	Methods	BJRoad		Shenzhen Road		FD Dataset		RS-RGB-T		VALID		Dongying	
		APLS	TOPO-F1	APLS	TOPO-F1	APLS	TOPO-F1	APLS	TOPO-F1	APLS	TOPO-F1	APLS	TOPO-F1
UniModal	UNet	52.1	58.7	56.2	51.0	60.6	65.7	74.5	70.2	69.4	64.0	77.1	72.3
	OARENet	55.9	61.4	58.7	53.1	63.2	67.9	77.2	72.4	71.8	67.3	79.9	75.0
	C^2Net	58.7	64.8	61.0	56.0	65.1	69.7	79.8	75.1	74.6	69.2	82.4	77.6
	RoadCT	61.0	66.3	62.9	56.8	66.6	71.4	81.5	76.7	75.9	71.1	83.8	79.0
	Road-SAM	64.7	71.9	66.7	60.9	68.5	73.2	83.3	78.6	79.2	74.7	87.0	81.8
Multimodal	DF-BSFNet	68.9	75.8	71.7	67.2	71.3	76.4	87.2	82.2	84.6	79.1	91.2	86.1
	DECCFNet	69.5	77.0	72.8	68.6	71.6	76.7	87.7	82.9	85.1	79.7	91.5	86.5
	CMIPNet	70.1	77.9	72.6	69.0	72.0	77.4	88.1	83.2	85.7	80.0	92.0	87.0
	LCIRE-Net	70.7	78.2	73.1	69.2	72.3	77.6	88.5	83.7	86.2	80.5	92.5	87.5
	MFNet	70.9	78.0	73.6	69.4	72.5	78.2	88.7	83.8	86.5	80.9	92.9	87.7
	FlexiSAM	71.7	78.9	74.4	70.2	73.4	78.7	89.2	84.2	87.0	81.2	93.5	88.2
	FTransUNet	71.9	79.2	74.3	71.1	73.6	79.3	89.4	84.8	87.2	81.7	93.7	88.9
	MultiSenseSeg	71.2	78.5	73.9	70.7	73.1	78.5	89.0	84.1	86.9	71.1	93.0	88.1
	SACFNet	74.8	80.1	75.6	71.8	75.8	82.1	90.2	85.7	88.6	82.8	94.4	89.5
	DEFNet	73.1	82.4	77.8	72.4	74.2	80.7	90.8	86.2	90.5	83.2	95.0	90.2
	RingMo-SAM	72.6	80.7	76.2	73.2	74.0	80.1	92.1	87.5	89.1	84.8	97.1	92.4
	AutoRoadSAM	76.2	83.8	79.1	75.0	77.2	83.7	93.3	88.4	91.8	86.0	98.0	93.6

TABLE III

QUANTITATIVE COMPARISON OF AUTOROADSAM AND 16 STATE-OF-THE-ART METHODS ON THE RS-RGB-T, VALID AND DONGYING DATASETS IN TERMS OF OA, m$F1$, MIOU, AND KAPPA. RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modality	Methods	RS-RGB-T				VALID				Dongying			
		OA	mF1	mIoU	Kappa	OA	mF1	mIoU	Kappa	OA	mF1	mIoU	Kappa
UniModal	UNet	82.41	76.20	73.18	79.60	74.26	69.50	65.56	70.18	79.43	74.18	72.56	76.84
	OARENet	85.70	77.92	77.66	78.10	73.76	68.34	67.92	70.35	85.14	82.39	78.45	82.63
	C^2Net	88.70	84.10	77.42	84.90	79.02	73.16	66.56	77.71	88.07	85.67	83.51	87.67
	RoadCT	85.47	80.23	75.12	83.45	77.34	71.67	68.19	74.82	87.56	84.73	82.41	86.28
	Road-SAM	89.89	82.43	80.06	88.92	81.74	74.41	68.81	76.95	90.06	86.37	84.44	89.10
Multimodal	DF-BSFNet	96.93	90.48	86.19	96.79	86.44	80.27	73.04	81.72	96.71	93.96	90.21	94.84
	DECCFNet	96.54	89.88	85.74	90.84	87.14	79.07	74.01	84.10	96.68	94.54	89.29	95.64
	CMIPNet	93.57	90.77	85.42	93.21	87.68	84.68	74.66	87.18	94.09	90.58	90.31	93.93
	LCIRE-Net	96.20	94.29	88.22	94.88	87.06	82.10	78.96	83.83	98.40	95.94	90.11	97.28
	MFNet	94.50	89.93	83.19	90.91	82.74	82.28	78.73	82.70	98.91	96.45	95.67	97.57
	FlexiSAM	94.59	90.22	87.48	91.28	84.98	80.13	76.32	84.28	98.99	96.16	93.09	98.14
	FTransUNet	96.12	93.47	86.91	94.83	86.74	82.16	78.53	84.92	97.65	95.28	94.73	96.81
	MultiSenseSeg	95.37	91.41	87.51	94.18	85.13	83.44	76.77	84.38	97.18	96.08	95.06	96.62
	SACFNet	95.72	92.84	86.61	93.29	86.75	83.07	79.45	85.12	98.41	95.73	95.12	97.54
	DEFNet	97.82	92.47	87.23	94.16	86.57	82.91	78.89	84.87	98.23	95.44	94.27	96.31
	RingMo-SAM	94.19	93.08	86.58	93.50	86.63	84.83	79.70	85.42	98.38	93.56	93.00	97.68
	AutoRoadSAM	98.65	93.13	90.25	96.82	89.13	85.33	81.21	86.33	98.08	97.09	96.89	97.37

B. Implementation Details

All experiments are conducted on workstations, equipped with an Intel Xeon Gold 6133 CPU (2.50 GHz), 24-GB RAM, and an NVIDIA GeForce RTX 3090 GPU, running Ubuntu 18.04. The model is implemented using PyTorch 2.0.0. The AdamW optimizer is employed for network training due to its effectiveness in mitigating overfitting and alleviating gradient-related issues. Following the hyperparameter settings of the original SAM framework and additional grid search optimization, the model is trained for 50 epochs. The initial learning rate and weight decay are set to 0.0001 and 0.01, respectively. To accommodate GPU memory constraints, the batch size is set to 1. All reported results are averaged over three independent runs to ensure reproducibility and robustness.

C. Evaluation Metrics

The performance of road segmentation is quantitatively assessed using overall accuracy (OA), mean $F1$-score (m$F1$), mean Intersection over Union (mIoU), Cohen's kappa coefficient (κ), and two widely adopted structure-aware metrics: average path length similarity (APLS) and topological correctness (TOPO). Let P denote the total number of pixels, K the number of classes, and for each class c, TP_c, FP_c, and FN_c represent the numbers of true positives, false positives, and false negatives, respectively. The OA is defined as the proportion of correctly classified pixels

$$\mathrm{OA} = \frac{1}{P}\sum_{p=1}^{P}\mathbb{I}\left(\hat{y}_p = y_p\right) \tag{32}$$

where $\hat{y}_p$ and y_p denote the predicted and ground-truth label for pixel p, and $\mathbb{I}(\cdot)$ is the indicator function. The $F1$-score and IoU for each class are calculated as

$$F1_c = \frac{2\mathrm{TP}_c}{2\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c} \tag{33}$$

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c} \tag{34}$$

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with the mean scores calculated by averaging over all classes

$$\mathrm{m}F1 = \frac{1}{K}\sum_{c=1}^{K} F1_c, \quad \mathrm{mIoU} = \frac{1}{K}\sum_{c=1}^{K} \mathrm{IoU}_c. \tag{35}$$

Cohen's kappa coefficient is used to measure the agreement between predictions and ground truth, defined as

$$\kappa = \frac{\mathrm{OA} - \sum_{c=1}^{K}\left(\frac{\mathrm{TP}_c+\mathrm{FP}_c}{P}\cdot\frac{\mathrm{TP}_c+\mathrm{FN}_c}{P}\right)}{1 - \sum_{c=1}^{K}\left(\frac{\mathrm{TP}_c+\mathrm{FP}_c}{P}\cdot\frac{\mathrm{TP}_c+\mathrm{FN}_c}{P}\right)} \tag{36}$$

where p_e is the expected agreement by chance. To evaluate structural integrity, we use two structure-aware metrics. APLS measures the similarity of shortest path lengths between node pairs S on predicted and ground-truth road graphs

$$\mathrm{APLS} = 1 - \frac{1}{|S|}\sum_{(i,j)\in S}\frac{\left|\ell_{\mathrm{pred}}(i,j) - \ell_{\mathrm{gt}}(i,j)\right|}{\ell_{\mathrm{gt}}(i,j)} \tag{37}$$

where $\ell_{\mathrm{pred}}(i,j)$ and $\ell_{\mathrm{gt}}(i,j)$ are the shortest path lengths between nodes i and j in the predicted and ground-truth graphs, respectively. TOPO is a graph-based metric that evaluates the topological similarity between predicted and ground-truth road networks. It samples seed vertices and compares the subgraphs within a fixed radius, computing precision (TOPO-P), recall (TOPO-R), and $F1$ score (TOPO-$F1$) as

$$\text{TOPO-P} = \frac{V_M}{V_P}, \quad \text{TOPO-R} = \frac{V_M}{V_G} \tag{38}$$

$$\text{TOPO-}F1 = \frac{2\times\text{TOPO-P}\times\text{TOPO-R}}{\text{TOPO-P}+\text{TOPO-R}} \tag{39}$$

where V_M is the number of matched vertices, V_P is the number in the predicted subgraph, and V_G in the ground truth. The TOPO-$F1$ score provides a comprehensive measure of both spatial and structural alignment between the networks.

D. Comparison With State-of-the-Art Methods

To comprehensively demonstrate the effectiveness of the proposed AutoRoadSAM framework, we compare its performance with 16 state-of-the-art semantic segmentation methods, including UNet [51], OARENet [38], C^{2}Net [37], Road-SAM [46], RoadCT [58], DF-BSFNet [59], DECCFNet [40], CMIPNet [43], LCIRE-Net [42], MFNet [52], FlexiSAM [60], FTransUNet [61], MultiSenseSeg [62], SACFNet [63], DEFNet [44], and RingMo-SAM [64]. All methods are evaluated on six benchmarks under identical experimental settings. The specific analysis is as follows.

1) Results on BJRoad, Shenzhen Road, and FD Dataset: As summarized in Table I, we compare AutoRoadSAM with sixteen state-of-the-art methods on the BJRoad, Shenzhen Road, and FD datasets, which cover two major modality combinations: OPT-Traj and OPT-DSM. Across all three datasets, AutoRoadSAM consistently achieves the best performance on the four evaluation metrics (OA, m$F1$, mIoU, and κ). On the BJRoad dataset, AutoRoadSAM achieves 85.64% (OA), 77.29% (m$F1$), 67.83% (mIoU), and 81.23% (κ), outperforming the second-best method by 1.97% and 1.66% in m$F1$ and mIoU, respectively. On the Shenzhen Road dataset, our model attains 73.10% (OA), 66.33% (m$F1$), 59.72% (mIoU), and 69.10% (κ), with improvements of 1.36% in OA and 2.21% in mIoU over the next best approach. These results indicate that AutoRoadSAM exhibits strong generalization capability for the OPT-Traj modality across diverse urban scenarios. On the FD dataset, which utilizes the OPT-DSM modality, AutoRoadSAM achieves 95.03% (OA), 91.29% (m$F1$), 87.81% (mIoU), and 93.12% (κ), surpassing the second-best method by 1.29%, 1.80%, and 2.03% in m$F1$, mIoU, and κ, respectively. This further demonstrates the robustness of our model in handling DSM-based multimodal information. Notably, AutoRoadSAM outperforms both recent unimodal and multimodal road extraction methods, as well as the latest SAM-based segmentation approaches. The superior performance can be attributed to the effective local alignment and global modeling of multimodal features via the LCAII module, which enhances both the discriminative representation of road objects and contextual feature fusion. In addition, as shown in Table II, AutoRoadSAM consistently achieves the highest APLS and TOPO scores on all three datasets, further demonstrating its superiority in capturing the structural connectivity and TOPO of road networks. Fig. 4 presents the qualitative results on representative scenes from the three datasets. In BJRoad and Shenzhen Road, where road objects often exhibit complex topologies and irregular boundaries, AutoRoadSAM achieves the most accurate and continuous road segmentation, effectively addressing boundary ambiguity issues that challenge other methods. In the FD dataset, despite subtle depth cues and less prominent road features, our model accurately decodes and fuses DSM and optical information to achieve precise extraction. Collectively, both quantitative and qualitative results demonstrate the strong generalization and adaptability of AutoRoadSAM across different modalities, scenes, and challenging road extraction scenarios.

2) Results on RS-RGB-T, VALID, and Dongying Dataset: As presented in Table III, we further evaluate AutoRoadSAM and sixteen comparative methods on three challenging multimodal road extraction datasets: RS-RGB-T, VALID, and Dongying. These datasets include three modality combinations (OPT-TIR, OPT-DSM, and OPT-SAR) and are derived from semantic segmentation sources, where abundant non-road objects such as vehicles, pedestrians, and trees increase the complexity of road extraction. AutoRoadSAM consistently achieves the best performance across all metrics and datasets. On RS-RGB-T, our method attains 98.65% (OA), 93.13% (m$F1$), 90.25% (mIoU), and 96.82% (κ), surpassing the second-best by 0.83%, 2.74%, and 1.94% in OA, mIoU, and κ, respectively. On the VALID dataset, AutoRoadSAM achieves 89.13%, 85.33%, 81.21%, and 86.33% for OA, m$F1$, mIoU, and κ, outperforming the runner-up by 1.45%, 0.5%, and 1.51% in OA, m$F1$, and mIoU. Despite the fragmented and imbalanced road objects in RS-RGB-T and VALID, our method delivers robust segmentation results, primarily owing to the local constraint strategy in the LCAII module, which effectively alleviates local discrepancies and enhances global multimodal feature fusion. On the Dongying dataset, AutoRoadSAM achieves 98.08% (OA), 97.09% (m$F1$), 96.89% (mIoU), and 97.37% (κ), with improvements of 0.64% and 1.22% in m$F1$ and mIoU over the second-best

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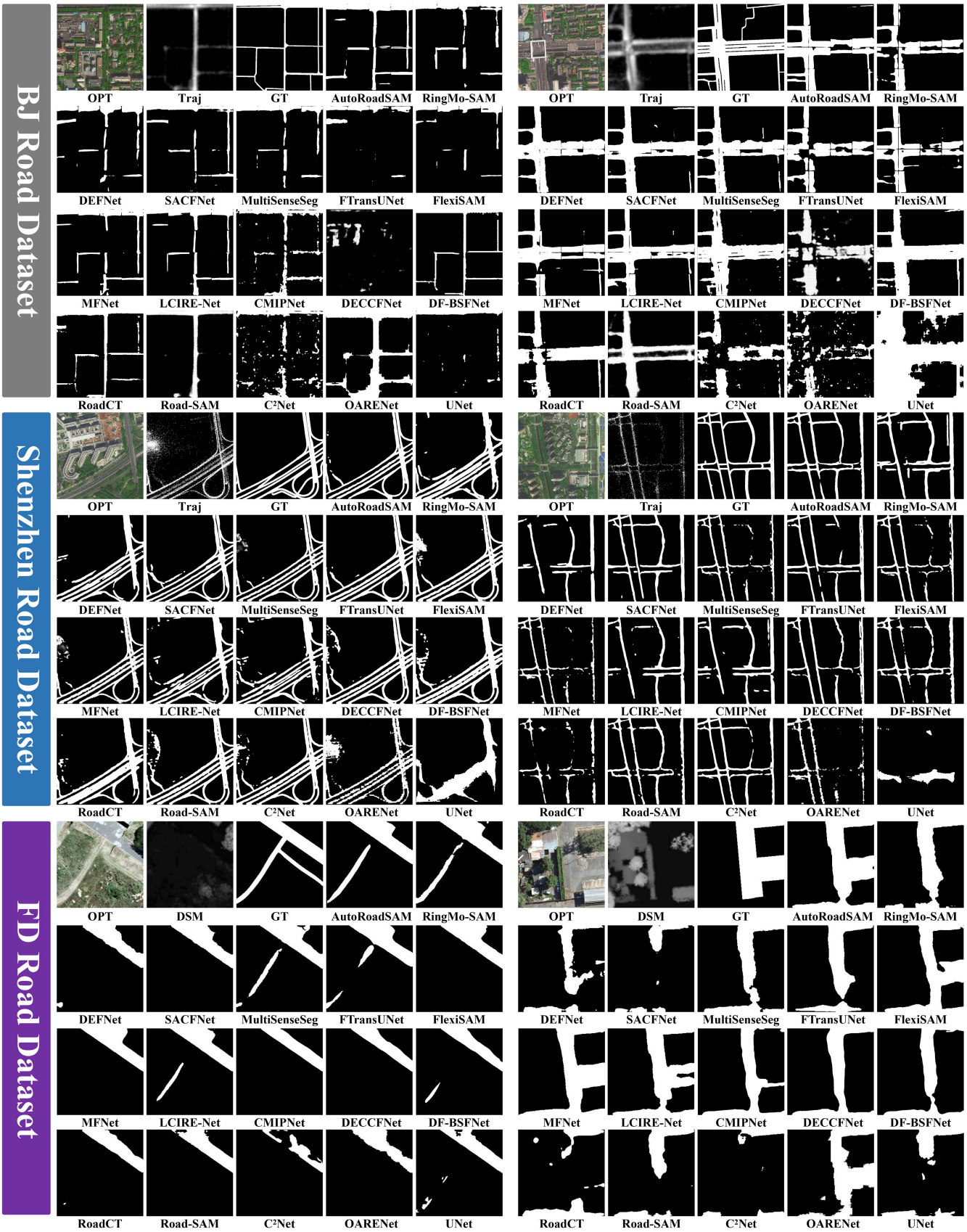


Fig. 4. Qualitative comparison of road segmentation results produced by AutoRoadSAM and 16 competing methods on representative scenes from the BJRoad, Shenzhen Road, and FD datasets.

approach. These results demonstrate the strong modality adaptability of our model, which extends effectively to SAR and TIR data. Moreover, as presented in Table II, our method outperforms all competitors in both APLS and TOPO metrics

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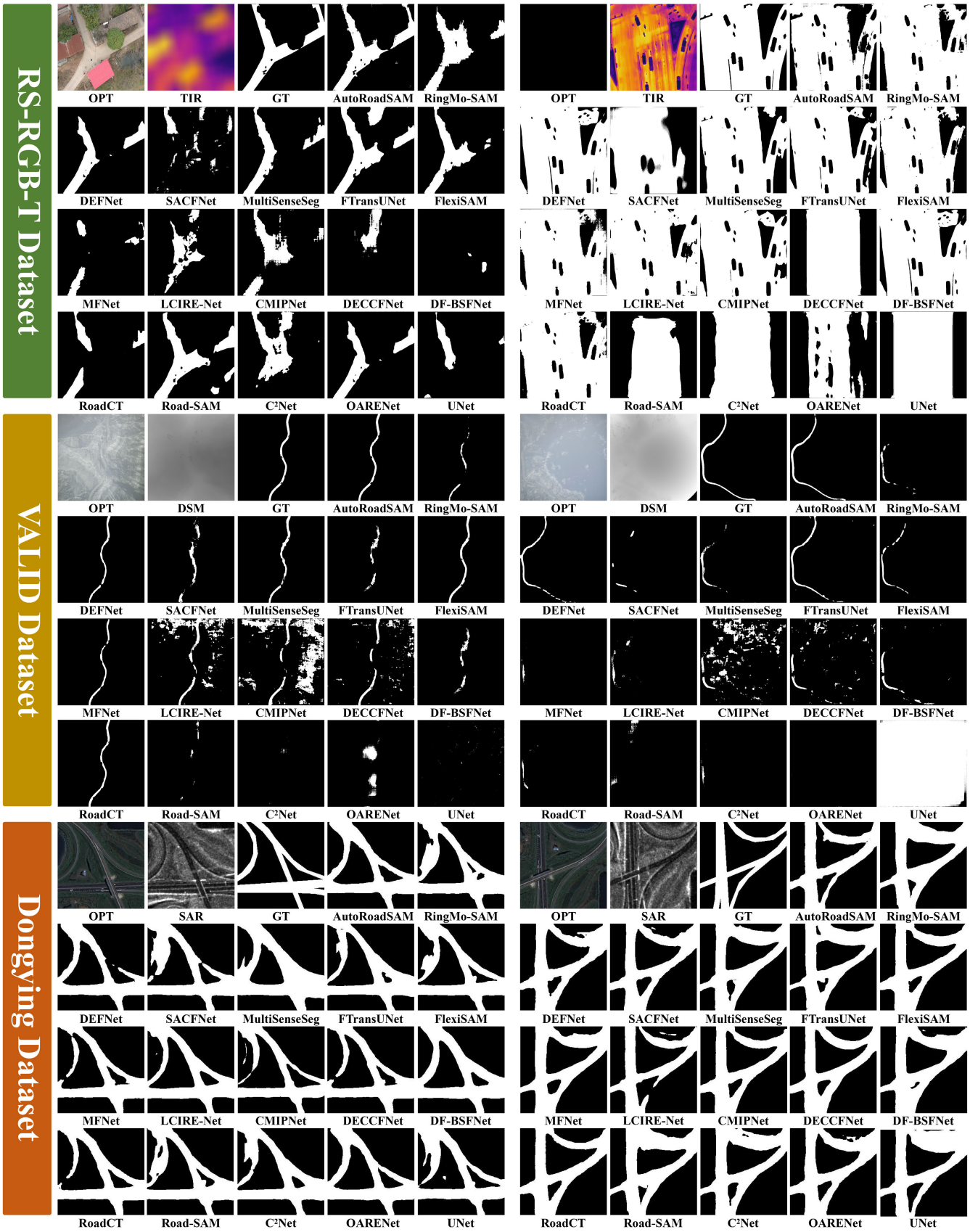


Fig. 5. Qualitative comparison of road segmentation results produced by AutoRoadSAM and 16 competing methods on representative scenes from the RS-RGB-T, VALID, and Dongying datasets.

across RS-RGB-T, VALID, and Dongying, highlighting its effectiveness in preserving road connectivity and topological structure under challenging multimodal conditions. Fig. 5 depicts qualitative comparisons on representative scenarios

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TABLE IV

COMPUTATIONAL COMPLEXITY ANALYSIS OF AUTOROADSAM AND REPRESENTATIVE UNIMODAL AND MULTIMODAL BASELINE METHODS ON THE VALID DATASET (INPUT SIZE: 1024×1024). RESULTS ARE REPORTED IN TERMS OF FLOPS, PARAMETER COUNT, MEMORY USAGE, INFERENCE SPEED (FPS), AND MIOU

Modality	Method	FLOPs (G)	Parameter (M)	Memory (MB)	Speed (FPS)	mIoU (%)
UniModal	UNet	120.8	62.5	2280	82	65.56
	OARENet	111.5	62.1	2390	78	67.92
	C^2Net	94.5	68.2	2511	69	66.56
	RoadCT	98.5	64.9	2455	72	68.19
	Road-SAM	93.2	70.4	2588	62	68.81
Multimodal	DF-BSFNet	85.2	82.1	2954	52	73.04
	DECCFNet	90.1	78.3	2701	55	74.01
	CMIPNet	46.6	130.1	4108	28	74.66
	LCIRE-Net	86.9	64.5	2512	62	78.96
	MFNet	85.1	66.1	2669	56	78.73
	FlexiSAM	50.8	124.2	4169	30	76.32
	FTransUNet	48.5	102.8	3571	33	78.53
	MultiSenseSeg	53.0	94.8	3169	32	76.77
	SACFNet	83.4	69.3	2418	49	79.45
	DEFNet	72.4	80.6	2890	48	78.89
	RingMo-SAM	58.3	108.4	3423	35	80.11
	AutoRoadSAM	65.7	89.5	3037	41	81.21

from the three datasets. In VALID, the severe class imbalance leads to poor performance of unimodal methods, whereas our CPLAM allows AutoRoadSAM to accurately segment roads despite dominant background pixels. For RS-RGB-T, the model robustly addresses modality-specific information loss, such as thermal degradation during the day and optical degradation at night, benefiting from adaptive dynamic fusion in the CMII module. In the OPT-SAR scenes from Dongying, substantial domain gaps and noise are present, yet our cross-attention-based mask decoder enables precise object recognition by deeply integrating prompt and fused features. These results confirm the superior generalization and multimodal adaptability of AutoRoadSAM under diverse and challenging road extraction conditions.

E. Complexity Comparison

To provide a comprehensive assessment of model efficiency, we evaluate the computational complexity of AutoRoadSAM and representative baseline methods on the VALID dataset. Table IV reports the FLOPs, parameter count, memory usage, and inference speed (FPS) for each method using 1024×1024 input images. AutoRoadSAM demonstrates a clear advantage in memory efficiency among multimodal models, with a memory usage of 3037 MB. This improvement is mainly attributed to the adoption of linear attention and the lightweight VMamba framework, both of which significantly reduce the model's memory footprint. While memory consumption remains higher than that of unimodal approaches, it is acceptable given the substantial accuracy gains delivered by AutoRoadSAM. In terms of inference speed, AutoRoadSAM is relatively slower than some existing methods. This is primarily due to the cascaded processing in the auto-prompting mechanism, where prompt generation depends on pseudo-label production, as well as the computational demands of the local constraint and contextual interaction modules within the LCAII module. Despite this, our model achieves the highest accuracy among all compared methods, striking a favorable balance between complexity and segmentation performance.

F. Ablation Studies

To comprehensively validate the effectiveness and rationality of our proposed architecture, we conduct extensive ablation studies on benchmark datasets.

1) Effectiveness of Each Module: To evaluate the effectiveness of the key modules in AutoRoadSAM, we conducted comprehensive additive and subtractive ablation studies on four representative datasets, Shenzhen Road, FD Road, RS-RGB-T, and Dongying Dataset, covering four different modalities. The original SAM with built-in box prompts serves as the baseline for all experiments. In the additive ablation experiments (Table V), we sequentially introduced the SAPD, the mask decoder with CPLAM (replacing the original mask decoder), and the CMII (replacing the original fusion module). With the gradual addition of these modules, the baseline model exhibited consistent improvements across all datasets. After integrating all components, the mIoU increased by 6.12%, 6.40%, 4.67%, and 3.46% on Shenzhen Road, FD Road, RS-RGB-T, and Dongying Dataset, respectively. These results highlight the limitations of simple box prompts in representing complex road structures and demonstrate that SAPD, through automatic point and mask prompts, significantly enhances the quality and informativeness of prompts. Notably, CMII and SAPD contribute comparable performance gains, while CPLAM yields a more modest improvement. This suggests that while SAPD diversifies prompt generation, CMII is particularly effective at capturing and integrating global and local multimodal features. In the subtractive ablation experiments (Table VI), we removed each key module (SAPD, CPLAM, or CMII) from AutoRoadSAM, one at a time, while keeping the other modules unchanged. The results show that removing any single module leads to a decline in segmentation performance across all evaluation metrics, confirming the necessity and complementarity of each module. Furthermore, the combination of SAPD and CMII demonstrates better synergy than other pairings. Fig. 6 provides a qualitative visualization of feature heatmaps after the sequential addition of each module. The attention maps indicate that, as modules are introduced,

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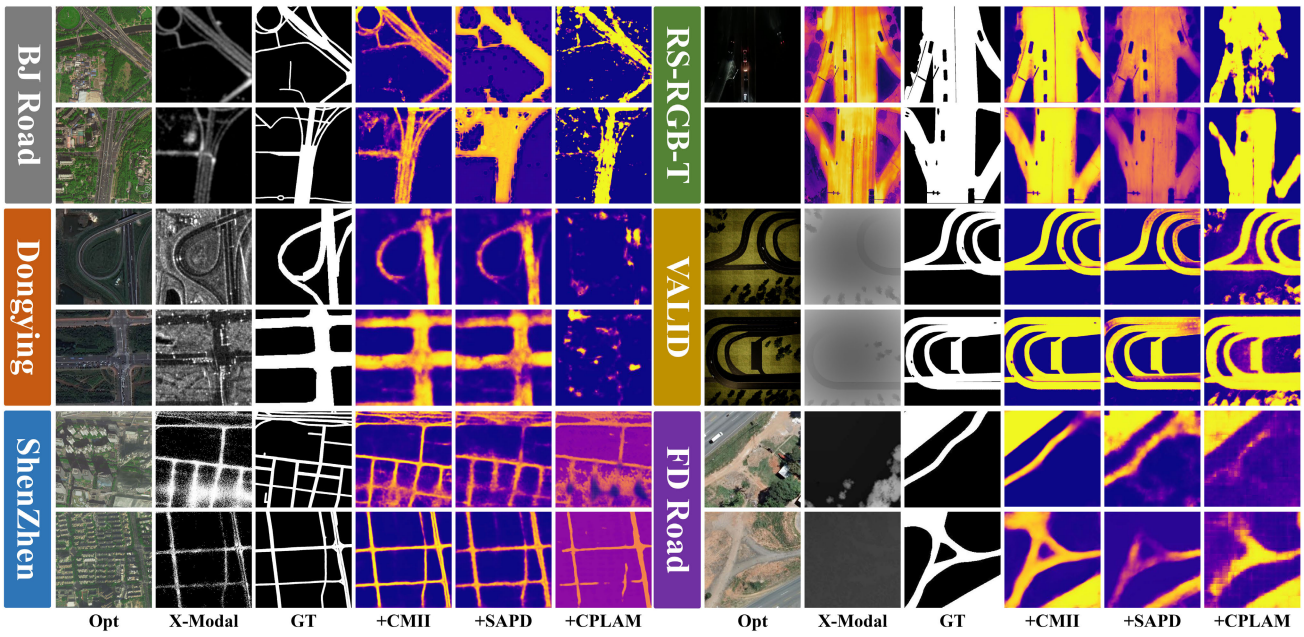


Fig. 6. Visualization of feature heatmaps after incorporating each module in the ablation experiments across six datasets. Brighter colors indicate stronger model attention, demonstrating the progressive improvement in road segmentation performance with each module.

TABLE V

ADDITIVE ABLATION STUDY ON DONGYING, RS-RGB-T, FD DATASET, AND SHENZHEN ROAD DATASETS. MODULES ARE INCREMENTALLY ADDED TO THE BASELINE TO EVALUATE THEIR INDIVIDUAL CONTRIBUTIONS. RESULTS ARE REPORTED FOR OA, M$F1$, AND MIOU

Model	CPLAM	SAPD	CMII	Dongying			RS-RGB-T			FD Dataset			Shenzhen Road		
				OA	mF1	mIoU	OA	mF1	mIoU	OA	mF1	mIoU	OA	mF1	mIoU
AutoRoadSAM	×	×	×	92.32	91.57	90.77	92.71	89.3	85.58	89.43	85.09	83.63	68.27	61.56	56.26
	✓	×	×	95.32	94.56	93.97	95.56	91.89	87.37	91.5	87.84	85.80	71.05	63.87	57.84
	✓	✓	×	97.06	95.6	95.36	96.62	93.19	88.00	92.51	88.86	86.44	71.84	64.80	58.51
	✓	✓	✓	98.08	97.09	96.89	98.65	93.13	90.25	95.03	91.29	87.81	73.10	66.33	59.72

TABLE VI

SUBTRACTIVE ABLATION STUDY ON DONGYING, RS-RGB-T, FD DATASET, AND SHENZHEN ROAD DATASETS. EACH KEY MODULE IS REMOVED FROM AUTOROADSAM IN TURN TO ASSESS ITS IMPACT ON SEGMENTATION PERFORMANCE. RESULTS ARE REPORTED FOR OA, M$F1$, AND MIOU

Model	CPLAM	SAPD	CMII	Dongying			RS-RGB-T			FD Dataset			Shenzhen Road		
				OA	mF1	mIoU	OA	mF1	mIoU	OA	mF1	mIoU	OA	mF1	mIoU
AutoRoadSAM	×	✓	✓	97.07	96.15	95.69	97.35	93.99	89.00	93.25	89.19	86.13	72.01	65.24	58.05
	✓	×	✓	96.80	95.40	94.16	94.70	91.69	87.54	93.13	86.68	85.18	70.91	65.00	57.33
	✓	✓	×	97.06	95.60	95.36	96.62	93.19	88.00	92.51	88.86	86.44	71.84	64.80	58.51
	✓	✓	✓	98.08	97.09	96.89	98.65	93.13	90.25	95.03	91.29	87.81	73.10	66.33	59.72

the model's focus on road regions becomes increasingly pronounced, reflecting progressive improvements in segmentation accuracy. SAPD is particularly effective at capturing global road contours, while CMII further refines boundary localization and enhances the integration of positional information. These quantitative and qualitative ablation results confirm the individual effectiveness and collaborative benefits of SAPD, CPLAM, and CMII within the AutoRoadSAM framework.

2) Effectiveness of Different Modal Data: To investigate the impact of auxiliary modality features on the main modality, we conducted an ablation study by removing the auxiliary branch in the feature extraction stage of AutoRoadSAM. As shown in Table VII, experiments are performed on four datasets, Shenzhen Road, FD Road, RS-RGB-T, and Dongying Dataset, each representing a different modality combination. The optical modality serves as the main branch in all modal combinations. In this setting, the CMII only employs the VMamba architecture and does not utilize SAPD, relying solely on externally provided bounding box prompts for segmentation. Experimental results indicate that the absence of auxiliary modality features leads to a noticeable decline in segmentation performance for the optical branch across all datasets. Specifically, the mIoU dropped by 2.89%, 4.51%, 1.84%, and 1.29% on Dongying, RS-RGB-T, FD Road, and

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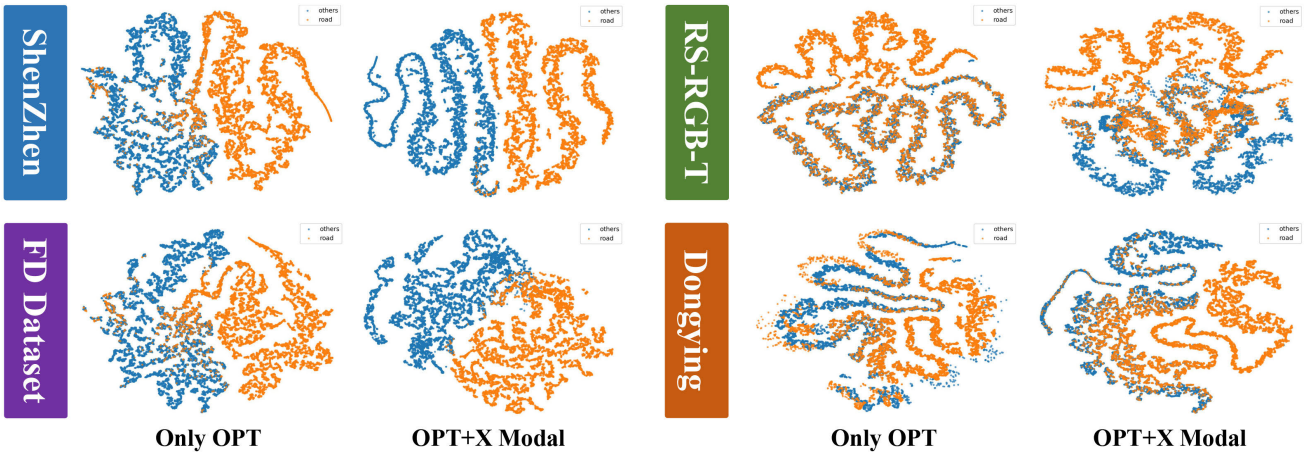


Fig. 7. t-SNE visualization of feature distributions for single-modal (only OPT) and multimodal (OPT+X Modal) inputs across four datasets. The addition of auxiliary modalities results in more compact and separable feature clusters, demonstrating improved road segmentation capability.

TABLE VII

ABLATION STUDY ON THE IMPACT OF AUXILIARY MODALITY FEATURES FOR DONGYING, RS-RGB-T, FD DATASET, AND SHENZHEN ROAD DATASETS. PERFORMANCE IS COMPARED BETWEEN USING ONLY THE OPTICAL BRANCH AND USING BOTH OPTICAL AND AUXILIARY MODALITIES. RESULTS ARE REPORTED IN TERMS OF M$F1$, OA, AND MIOU

Model	Optical	X Modal	Dongying			RS-RGB-T			FD Dataset			Shenzhen Road		
			mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU
AutoRoadSAM	✓	×	97.11	95.17	94.00	94.70	92.17	85.74	90.28	88.68	85.97	70.91	63.01	58.33
	✓	✓	98.08	97.09	96.89	98.65	93.13	90.25	95.03	91.29	87.81	73.10	66.33	59.72

Shenzhen Road, respectively. This demonstrates that auxiliary modalities contribute significant complementary information to the road extraction process, especially in OPT-TIR and OPT-SAR combinations. For example, in the RS-RGB-T dataset, a portion of the optical data is acquired at night, where the optical modality alone provides limited information. The addition of thermal or SAR data, which offer distinctive and effective features, substantially improves segmentation accuracy. Furthermore, as illustrated in Fig. 7, we present t-SNE visualizations of feature distributions for single-modal and multimodal inputs across the four datasets. The plots show that the introduction of auxiliary modalities enhances feature clustering and separability, further confirming the positive effect of multimodal fusion on road extraction.

V. CONCLUSION

To address the challenges of multimodal road extraction, such as complex geometric structures, topological continuity, and class imbalance, we propose AutoRoadSAM, an end-to-end vision-based segmentation framework based on the SAM. AutoRoadSAM introduces an auto-prompt decoding mechanism with snake convolution to eliminate dependence on external prompts, and incorporates an improved scanning strategy in SS2D for enhanced semantic continuity and topological representation. The feature extraction stage employs a lightweight VMamba backbone, augmented with local constraint and contextual fusion modules, to efficiently capture multimodal contextual information. Furthermore, a polarity-aware mask decoder is designed to mitigate the class imbalance problem by leveraging both positive and negative feature cues during decoding. Comprehensive experiments on six diverse datasets with four types of modalities demonstrate that AutoRoadSAM consistently outperforms state-of-the-art approaches in both accuracy and generalization. Despite these advances, real-time inference remains a challenge for the current model. Future research will focus on developing more efficient and lightweight multimodal foundation models for practical road extraction applications.

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